Auto-Grasp: will our fingers grip the new part?

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The problem

A new part is coming to a robot cell. **Can the fingers we already have grip it reliably, and if not, which finger should we make?** Today the answer comes from making fingers and trying them on the robot. That takes days, ties up the cell, and only tests the positions someone thought to try. No camera or robot grasps exactly where it aims, and a finger that works at the centre can drop the part a few millimetres off.

What we built

Auto-Grasp is a web app that answers that question before anything is printed. Its inputs are the part's STEP or STL file, the cell's gripper, and how accurately the cell grasps. It works **library first**:

1. **Rank the library.** Every finger the factory owns is simulated against the new part in NVIDIA Isaac Sim, at the part's candidate grasps and across the cell's grasp error, and ranked by predicted success.
2. **The engineer decides** whether the best finger is good enough. There's no fixed pass bar.
3. **Generate only if needed.** Bayesian optimisation (BoTorch) searches a printable finger family, several designs per simulation request. The winner is exported as STEP and STL with the gripper's mount holes and an engraved ID, passes a review gate, and joins the library.

How it scores a finger

- **Digital twin:** the organizers' FANUC ER-4iA cell and SMC MHZL2-16D2 gripper in Isaac Sim 6.0, on the MSOE NVIDIA GB10, running a top-down pick.
- **Error grid:** each finger is tried at 963 position and yaw offsets (out to 12 mm and 12°). Stored outcomes re-score at any grasp accuracy with no new simulation, which gives a success-vs-error curve.
- **Honest holds:** if a finger knocks the part before the jaws close, the grasp counts as a miss, even if the simulator still held it.

Why it matters for manufacturers

- Fewer custom fingers: most new parts run on fingers already on the shelf, and the app says which.
- Less time on the cell: 4 fingers ranked on a part in **10 min** of simulation; a new finger designed in **33 min**.
- A number to decide with, and the error it holds up to, instead of a guess.

Results, in simulation and on the real arm

Predicted success is at the cell's assumed grasp accuracy (3 mm and 8°, one standard deviation). Measured results are from the organizers' platform at CSI, with 95% Wilson intervals.

Part (task: pick and drop)	Finger	Predicted (Isaac)	Measured on the arm
16 mm cube, as wide as the existing fingers open	Existing hooked-tip fingers	0%	not run (doesn't fit)
	Best other library finger	17.6%	not run
	Generated finger (D1)	**92.3%**	**7/8 scored, 87.5% (53–98%)**
M6 × 30 screw, fits the existing fingers	**Existing hooked-tip fingers**	**79%**	4/10, 40% (17–69%)
	Generated finger (D1)	76%	not run
	Imprint of the part	69%	not run

- **The cube is where generation pays off.** The library can't grip it. The generated finger holds every simulated grasp out to 4.5 mm of error. It was found from 28 candidate designs and 12,105 simulated grasps, then printed, and on the real arm it held at every position offset out to 4 mm, as predicted. The one miss, at 12° of yaw, came with a controller fault.
- **On the screw, Auto-Grasp says not to print anything.** Its Ø6 shank sits in the existing fingers' sweet spot, and the gripper's 6 mm stroke per jaw caps any custom finger's slack. Every good generated design hit that cap, and none beat the existing fingers.

Limitations, where it fails

- **The screw scored lower on the arm than predicted** (40% against 79%). Contact with a small part lying on its side is noisy in simulation: repeated runs of the same fingers on the screw scored 63–79%, and about 1 offset in 5 flips between runs.
- **Unmeasured inputs:** the 3 mm / 8° grasp accuracy is a placeholder, printed-finger friction is estimated, and height error isn't simulated yet. The simulated grip is weak (about 2.5 N per jaw), fine for light parts only.
- **Small real-arm samples:** one session, 8 to 10 rollouts per finger. No printed screw finger has run yet.

Stack: Isaac Sim 6.0, BoTorch, CadQuery, FastAPI, React. Code, experiment logs and every number above: github.com/ALL-Applied-AI-Network/msoe-ai-club-rockwell-automation-nvidia-hackath-alec-weinbender-s-team-a715330bad